

“Turing proposed a class of devices that came to be known as Turing machines.

“The proposition that Turing’s notion (that class of devices) captures exactly the intuitive idea of effective procedure is called the Church-Turing thesis. This proposition, being a claim about the relationship between a formal concept and intuition, is not provable, though it would be refuted by an intuitively acceptable algorithm for a task that is not Turing-computable. That no such counterexample has been found, together with the fact that... indicates that there is at least something natural about this notion of computability.

“Turing machines are not physical objects but mathematical ones. The architecture is simply described, and the actions that may be carried out by the machine are simple and unambiguously specified. Turing recognized that it is not necessary to talk about how the machine carries out its actions, but merely to take as given the twin ideas that the machine can carry out the specified actions, and that those actions may be uniquely described.”

The third paragraph might warrant explanation: what Turing proposed as his TM is the same as the (intuitive) idea that something *can* be computed. Now *this* cannot be proved right, of course, but still, if one were *able* to find an algorithm for a task that Turing said was *not* computable, then that algorithm would prove it *wrong*.

Anyone—absolutely anyone—interested in how our computers are Turing machines and how they came to be should visit <http://plato.stanford.edu/entries/turing-machine/>, which supplies an excellent introduction.

## 1937

George Stibitz of Bell Labs constructed a 1-bit ~binary~ adder using relays. This was, therefore, one of the first binary computers, although it was only a demonstration. Improvements led to the Complex Number Calculator of 1940.